Week 3 web page: delivery guide

Instructor notes for Week3_ProbabilityAndBayes.html

General

- Page: https://kevintshoemaker.github.io/NRES-746/Week3_ProbabilityAndBayes.html. Open it before class; the three widgets need no internet once loaded.
- For live demos, run the matching section of `CODE_Week3_ProbabilityAndBayes.R` (same seeds, same numbers). Everything runs in a second.
- Zoom the browser to 125-150% for the projector.
- Questions on the page have no answers. Answers are below; *Land:* is where discussion should end up.
- Readings for Canvas: Bolker 4.1-4.3 (printed pp. 139-152).
- Labor Day: no Monday class, so Lab 3 (Tuesday) came *before* the only lecture. The lab H1 is the short opening for the lab; the Wednesday H1 names and formalizes what Lab 3 Act 1 had students work out by counting.
- WOD: *A Probability Puzzle* (key in `wod/answer_keys/`). It was not reached Wednesday and ran the next Monday; the page points to its worked version on the Week 4 page (`#wod-puzzle`).

As taught (Sept. 8-9, 2026)

- **Tuesday (Lab 3):** short opening on the addition and multiplication rules, then three pen-and-paper Acts (no laptops): Act 1 derived the PPV of a camera-trap classifier from a natural-frequency table, Act 2 compared binomial and Poisson detection counts, Act 3 read a normal density and CDF and used a printed z-table. Per the announcement (added Nov. 4): students attempted to derive Bayes' rule from scratch (with mixed success); unfinished Act 3s were handed in the next day or later. The announcement also set up the Week 4 project pitches; its eight pitch elements and the pitch/vote process are now on `FINALPROJ.html#pitch`, and its "don't worry, keep rereading ch. 4" note closes the page's `#lab3` part.
- **Wednesday:** the probability lecture (axioms, the two-way table, conditional probability, Bayes' rule with Bolker's medical test), following the PDF NOTES now in `lecture_notes/archive/`. The WOD was not reached. Per the announcement (added Nov. 4), making Bayes' rule feel intuitive didn't land; the announcement pointed students to 3Blue1Brown's Bayes theorem lesson, now linked in `#bayes`.
- The page was built Oct. 30, after the week, from those NOTES, the archived `NOTES_Lab3`, legacy `LECTURE2` (now in `archive/`; its distributions half is on the Week

4 page), and Bolker 4.1-4.3. New on the page: the three widgets, the seed-fate and double-observer examples, the tree plot (LECTURE2's urn with the same counts), the discussion prompts, and both bonuses.

Part 1: Adding and multiplying probabilities (lab opening, ~5-8 min)

- Keep it short. Two rules unlock Act 1: add probabilities of mutually exclusive outcomes; a probability times a count gives an expected count. Don't use $P(A \mid B)$ notation or derive Bayes' rule before the Acts; Act 1 gets there by counting, and Wednesday names it.
- Seed fates: $P(\text{eaten}) = 0.35 + 0.15 = 0.50$; complement $1 - 0.10 = 0.90$. 200 seeds $\times$ $0.35 = 70$ expected.
- Double observer: 0.36 / 0.24 / 0.24 / 0.16; at least one 0.84. Ask why $0.6 + 0.6 = 1.2$ fails before showing the general addition rule.
- "Real nests differ in how well they are hidden, so a nest one observer misses is often missed by the other too. Is the true chance that both miss a nest more or less than 0.16? What does that do to an estimate of how many nests nobody found?" *Land:* more than 0.16; misses are positively correlated. An estimate that assumes independence says 16% are missed, so it undercounts the nests nobody found (heterogeneous detectability biases double-observer and mark-recapture estimates low).
- Act keys: `labs/answer_keys/Lab3_Act1_Key`, `_Act2_Key`, `_Act3_Key`. Quick checks while circulating: Act 1 PPV $\approx$ 1% at the rare site, $\approx$ 50% at the common one (a PPV above 0.5 at the rare site means the fraction is upside down); Act 2 $P(X = 3) = 0.2541$, mode 2, variances 1.68 vs. 2.4 and 2.34 vs. 2.4; Act 3 $z = 1.67$, $P(\text{mass} > 5.5) \approx 0.048$, top 10% above 5.27 kg.

Part 2: Conditional probability and Bayes' rule (Wednesday, ~35 min)

Big picture (~2 min)

- Point at FIT. A model gives $P(\text{data} \mid \text{hypothesis})$; we want the reverse. Name the likelihood.
- Ask what recipe groups found in Act 1 and where they got stuck; the rest of the hour gives it notation.

Joint, marginal, conditional (~10 min)

- Tree table: 241 trees. Product of marginals 0.219 vs. joint 0.104: let someone propose multiplying first.
- $P(\text{healthy} \mid \text{pinyon}) = 25/126 = 0.198$; juniper 0.661. Conditioning throws out the other column and rescales.

- Law of total probability gives 0.419. Mention that Bolker's p. 142 version drops the weights.
- Widget: "Tree plot" first (cards: joint 0.104, $P(B)$ 0.419, $P(A \mid B)$ 0.248, product 0.219). Then "Make B independent of A."
- "Start from the tree plot settings and move $P(B \mid \text{not } A)$ until the top of the blue region is level across both columns. What happens to $P(A \mid B)$, and how does it compare with $P(A)$? What does a level line say about whether species tells you anything about infestation?" *Land:* level at 0.198, $P(A \mid B)$ becomes 0.523, equal to $P(A)$, and the joint equals the product. A level line is independence: knowing infestation status tells you nothing about species, and the reverse.

Bayes' rule (~12 min)

- **Board:** write $P(A \text{ and } B)$ both ways, divide by $P(B)$.
- Trees: $P(\text{healthy} \mid \text{pinyon}) = 0.198$ vs. $P(\text{pinyon} \mid \text{healthy}) = 0.248$.
- Medical test: $P(+) \approx 0.01$, PPV ≈ 0.0001. In counts: 100 million people, 100 infected positives vs. about a million false positives. Name the base rate fallacy, then connect back to Act 1.
- Widget defaults: prevalence 0.02, sensitivity 0.9, FPR 0.05 give PPV 0.269 (670 positives per 10,000, 180 real).
- "Move sensitivity across its full range, then put it back and do the same with the false positive rate. Which one matters more for $P(\text{present} \mid +)$, and why? What happens when the false positive rate is 0?" *Land:* sensitivity moves PPV only from 0.17 to 0.29; the FPR moves it from 1.0 down to 0.08. False positives come from the 98% of samples that are absent, so even a small FPR swamps the true positives. With FPR 0 every positive is real (PPV 1) at any prevalence. Prevalence matters as much as the FPR (0.018 to 0.947 across its range).

More than two hypotheses (~6 min)

- Liana: posterior 0.179, 0.179, 0.429, 0.214; heavy rises from 0.05 to 0.21. Likelihoods sum to 1.05: they aren't a distribution over classes.
- One sentence on Bayesian vs. frequentist: the dispute is about priors on parameters, not about Bayes' rule.
- "Suppose you knew nothing about how common each class is and gave all four the same prior, 0.25. What would the posterior for heavy infestation be? Why is it so different from 0.21?" *Land:* $0.6/1.05 = 0.571$. With equal priors the posterior is the likelihood rescaled. The real prior puts only 5% on heavy, and that base rate pulls the answer down: the same evidence, different starting points.

Discussion (~3-5 min)

- Q1 (xkcd sun detector) *Land:* posterior odds = prior odds $\times \frac{35/36}{1/36}$ = prior odds $\times$ 35. The posterior passes 0.5 only if the prior exceeds $1/36 \approx 0.028$. The real prior is astronomically small, so a "yes" is almost surely a lie. The p-value answers $P(\text{yes} \mid \text{no nova})$, not $P(\text{nova} \mid \text{yes})$.

- Q2 (80% of failed nests had a predator on camera) *Land:* the study measured $P(\text{predator} \mid \text{failed})$; the manager wants $P(\text{failed} \mid \text{predator})$. You also need the failure rate and how often successful nests show a predator. Example: 30% of nests fail and 40% of successful nests also show a predator gives $0.24/(0.24 + 0.28) = 0.46$, not 0.8.
- Q3 (two eDNA positives) *Land:* with the widget's defaults, one positive gives 0.27 and a second independent positive 0.87 (each multiplies the odds by $0.9/0.05 = 18$). Same-day samples share contamination sources (gear, crew, lab batch, DNA carried in by boats or birds), so false positives are correlated and the second positive adds less than that. Independent confirmation means a different day, crew, or lab.

Bonus: Monty Hall (skip if behind)

- Likelihoods 1/2, 0, 1; posterior 1/3 stay, 2/3 switch; simulation 0.336 and 0.662 (seed 31). Widget: 1,000 games at 3 doors, then 10 (switch wins 0.9). The host's knowledge is what makes the likelihoods unequal.

Bonus: zero inflation (skip if behind)

- $P(0) = 0.3 + 0.7 \times 0.6^5 = 0.354$ (Bolker Fig. 4.1, right). Two processes (visit or not; how many taken) give extra zeros.